Spillover effects within supply chains: Evidence from Chinese-listed firms

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Abstract: There is increasing attention on information transfers along supply chain partners for firm (extreme) events. This growing literature finds spillover effects following certain types of firm events. Using data from credit rating actions of Chinese-listed firms over the period between March 2007 and May 2020, we examine the spillover effects of supply chains by focusing on the market reactions of event firms to the action announcements. We find strong evidence of spillover effects driven by the market reactions of event firms, which are enhanced through information diffusion channels as supply chain partners receive more investor attention. Moreover, the effects are stronger when event firms' market reactions are negative, event firms are non-stated-owned, the industry concentration of event firms is higher, or the supplier-customer business relationship is closer. Overall, these findings highlight the role of investor attention in addition to network characteristics in supply chain spillovers.

Keywords: Spillover effects; Supply chains; Market reactions; Credit rating; Information diffusion

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1. Introduction

This study investigates the spillover effects of supply chains in China by focusing on the market reactions of firms to announcements of credit rating actions such as rating changes, credit watches and outlooks. While previous research has extensively explored the wealth effects of adverse extreme events such as bankruptcy filings and natural disasters (e.g., Hertzel et al. 2008; Houston et al. 2016; Kolay et al. 2016; Jacobson and von Schedvin 2015; Hendricks et al. 2020), there may be little exploration into whether spillover effects driven by stock market reactions of credit rating actions exist and, if so, what affects them. This study focuses on the idea that firms in the supply chain will experience comovements during credit rating events. We are concerned about the credit rating actions because they are more common, cover more firms and hence would have more comprehensive effects along the supply chain.

Indeed, Chang et al. (2015) recently explore the information effects of bond rating downgrades along the supply chains. They find that suppliers experience significant credit spread increases during customers' bond rating downgrades. This indicates that the announcement of credit rating downgrades has credit contagion effects on downgraded firms' suppliers. However, we focus on spillover effects in stock markets driven by market reactions of all credit rating actions. In addition to rating changes, credit rating agencies issue other rating actions, such as credit watches and rating outlooks, which also contain valuable information (Cantor and Mann 2007; Chung et al. 2012). Given the importance of credit watches and rating outlooks actions, it is reasonable to believe that these actions may have spillover effects in the supply chain. Moreover, there is scarce empirical evidence on spillover effects of credit rating actions in the supply chain of emerging economies, particularly in China, where the market for firm crediting ratings is large and growing. Credit ratings have become more important

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for Chinese firms since 2005 when the People's Bank of China issued the measures for the administration of short-term financing bonds. Credit rating is a reliable passport for firms to obtain government support and external financing, which is widely concerned by suppliers, customers and investors of firms. Thus, these facts emphasize the importance of our investigation.

Our focus on China's supply chain is driven by the unique characteristics of its firms, which differ from those in developed countries. One particular feature is that state-owned firms account for a large proportion in China's economic market.[1] Compared with non-state-owned firms, the state-owned firms with closer ties to the government are more likely to obtain scarce resources such as preferential loans and tax exemptions, which makes state-owned firms usually in a strong position in the market (Shleifer and Vishny 1994; Allen et al. 2005), and have relatively soft budget constraints (Cunat 2007). With these arguments in mind, it is important to explore the spillover effects of credit rating actions in the Chinese supply chains. Another important feature is the relatively lower level of industrial concentration among firms in China compared to developed countries. The industrial concentration affects the implications of spillover effects in the supply chain (Gosman et al. 2004; Hertzel et al. 2008). Furthermore, Chinese firms are involved in a wide range of industries, which is an important feature of the development of emerging economies. Credit rating events cover almost all industries, which may make our results of supply chain spillovers more representative.

To explore the existence of spillover effects among the firms in the supply chain, we examine the relationship between the abnormal returns of an event firm and the abnormal returns of the portfolio of suppliers and customers. We focus on the market

[1] By May 2020, the number of state-owned enterprises in the listed companies' accounts for about 28%, and their outstanding market value accounts for about 46%.

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reaction of an event firm rather than the event itself. The basic intuition behind these analyses is as follows: provided that supplier firm and customer firm are each other's stakeholders, the change of a firm's credit rating, especially the unexpected component of the change, often predicts the change of the performance and financial health of its suppliers and customers, thus generating spillover effects in the supply chain. More specifically, information about rating changes is exchanged not only between supply chain partners, but also among their investors. Rating changes, especially those are unanticipated, will cause market reactions under the market expectation theory (Smith 1986), and further lead to stock price reactions of other stakeholders under the hypothesis of investor attention (Hou et al. 2009). Thus, the abnormal market reaction to an event is more important than the event's characteristics with regard to spillovers in the supply chain. Above all, the market reaction conditions the market's expectations about the credit rating change (credit watch or outlook) event; therefore, the market reaction better captures the surprise component associated with rating changes (credit watches or outlooks) compared to the changes themselves. In other words, whether the market participants view the rating change (credit watch or outlook) of an event firm favorably or unfavorably is more meaningful than whether the rating change (credit watch or outlook) of the firm is upgraded or downgraded (positive or negative). Indeed, we find that the market evaluation of a firm by an event plays an important role in determining the characteristics of spillover effects in the supply chain.

The central finding of our study is that suppliers and customers as a whole experience significant spillover effects driven by the market reactions of event firms to credit rating actions such as rating changes, credit watches, and outlooks. This finding is robust to endogeneity, estimation approaches for abnormal returns, and event windows. This novel finding provides direct evidence that credit rating actions have a

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greater economy-wide impact, in addition to the previously documented effects only on the event firms. The spillover effects are economically significant. On average, a one standard deviation increase in the market reaction of an event firm is associated with a 1.65% increase in the CAR(0,4) of nonevent firms in the supply chain.

To explore the information channel of our documented spillover effects, we augment our specification with indicator variables for investor attention and find that this factor has a substantial effect. The spillover effects are 3.4 times larger if both supply chain partners are being followed by more of the same analysts. Meanwhile, the spillover effects are 1.4 times larger if both supply chain partners are being covered by more of the same media. Thus, the spillover effect becomes more pronounced when information about supply chains improves.

To drill further down, we conduct several cross-sectional analyses. Firstly, we extend our analyses by comparing negative spillovers with positive spillovers. We refer to negative (positive) spillovers as the spillover effects caused by negative (positive) market reactions of event firms. Our results show that negative spillovers are more dominant than positive spillovers, which are consistent with the argument that negative events generally have a larger impact on market participants (Tversky 1979; Soroka 2006). Secondly, we find that the spillover effects are larger when the event firms are non-state-owned. This result suggests that market participants are largely more concerned about the credit rating actions of private firms, because they are more dependent on private investments and banks loans for growth, while state-owned firms are more likely to have access to scarce resources (Cong et al. 2019). Thirdly, we examine the relationship between spillover effects and industry concentration. To that end, we use the Herfindahl ratio to measure the industry concentration of event firms, which is constructed from the sales of event firms for the whole industry. We find that

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the spillover effects are stronger when the industry concentration of event firms is higher.

Finally, we consider several alternative specifications based on the portfolio of suppliers or customers separately. We find that the spillover effects due to credit rating actions can spread not only upstream, but downstream as well. This result bolsters our evidence on individual suppliers and customers and constitutes new evidence on the extent of spillover effects of credit rating events. In addition, the spillover effects to suppliers appear to be greater than those to customers. Specifically, we observe that the spillover effects to suppliers rather than to customers are amplified when the rating change of even firms is across two grades (investment grades and speculative grades) or rating agencies have global partners. These results imply that rating changes can affect the bargaining powers between suppliers and customers in different ways. Moreover, we find that when the level of business relationship between event firms and their suppliers and customers is higher, the spillover effects to suppliers and customers are both stronger.

This paper makes several contributions to the existing literature. Firstly, to the best of our knowledge, this paper is the first to uncover the spillover effects driven by the stock market reaction of an event firm (to credit rating actions) in the supply chain. We focus on the abnormal market reaction of event firms rather than whether the events are favorable or unfavorable, which captures the surprise component of events better than the events themselves. We provide direct evidence that both suppliers and customers experience the spillover effects driven by the market reactions of event firms, which suggests that the market evaluation of a firm with regard to an event is crucial in order to determine the characteristics of spillover effects. Secondly, this paper examines the spillover effects of all credit rating actions including credit watches and outlooks in

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addition to rating changes. Many studies have focused on the impacts of credit rating changes, particularly downgrades (e.g., Weinstein 1977; Jorion et al. 2005; Jorion and Zhang 2010; Chang et al. 2015; Alldredge et al. 2019), but relatively few have looked at the impacts of credit watches and outlooks. Therefore, in a broader sense, our results show that credit rating action events have supply chain spillover effects. Thirdly, this study contributes to the literature on the role of attention by highlighting that the documented spillover effects are stronger if supply chain partners are being followed by more of the same analysts or covered by more of the same media. This finding improves our understanding of the propagation of firm level shocks through information flows along supply chains. Overall, this paper provides valuable insights into the spillover effects driven by market reactions of event firms to credit rating actions in the supply chain, considers a comprehensive range of credit rating actions, and sheds light on the role of attention in shaping spillover effects.

The rest of the paper is organized as follows. Section 2 presents a review of the related literature and develops the hypotheses. Section 3 describes our data and empirical approach. Section 4 presents the main empirical results. Section 5 conducts further analysis. Section 6 conducts robustness checks. Section 7 concludes the paper.

2. Related literature and hypothesis development

2.1. Related literature

Our paper is closely related to a number of papers on supply chains as channels for information in financial markets. This literature has focused on adverse events, such as bankruptcies, defaults, credit rating downgrades, and nature disasters (e.g., Hertzel et al. 2008; Houston et al. 2016; Jacobson and von Schedvin 2015; Kolay et al. 2016; Chang et al. 2015; Hendricks et al. 2020). In contrast,, our paper contributes to this

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literature by investigating both favorable and adverse shocks, and focusing on the market reactions of event firms rather than the events themselves. We find that the spillover effect driven by the market reactions of event firms is significantly positive, especially the negative spillover driven by negative market reactions is stronger. This contributes to the existing literature by highlighting the distinct dynamics of spillover effects resulting from market reactions.

Our paper is also relevant to the literature on information transfer effect of credit events on partners along the supply chain. A typical recent work by Chang et al. (2015) uses credit default swap (CDS) spread data from Markit and an event-study method to estimate credit risk effects of credit rating downgrades on downgraded firms' suppliers and customers. Chang et al. (2015) find that suppliers experience significant credit spread increases during rating downgrades. In parallel, Agca et al. (2022) define CDS jump events and analyze credit shocks propagate driven by these events through supply chains in the CDS market. They find that the CDS jump events result in substantial abnormal CDS spread changes for the supply chain partners of the event firm, in the same direction as the jumps. Besides rating changes, credit rating agencies also issue other rating actions such as credit watches and rating outlooks that contain important information, but are overlooked by the prior literature (e.g., Cohen and Frazzini 2008; Menzly and Ozbas 2010; Chang et al. 2015; Alldredge et al. 2019; Agca et al. 2022). Our paper contributes to this literature by examining the stock market spillover effects of credit rating actions through supply chains, including rating changes, credit watches, and outlooks. Specifically, we focus on the spillover effects driven by the stock market reaction of rating changes, credit watches and outlooks events' firms. We find that both suppliers and customers experience positive spillover effects, which remain robust across different types of rating actions.

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Moreover, our paper is connected to the literature that explores how network characteristics, such as network centrality (Wu and Birge 2014; Yang and Zhang 2019), customer concentration (Cen et al. 2017; Campello and Gao 2017), customer and supplier state ownerships (Allen et al. 2005; Cunat 2007), and customer relationships (Cen et al. 2016) affect the revenues and valuation of firms. The current paper contributes to the prior literature by focusing on the association between spillover effects in the supply chain driven by the market reaction of event firms (to credit rating actions) and network characteristics. We present evidence that the spillover effects in the supply chain are considerably stronger when event firms are non-state-owned. Besides, we find that the spillover effects in the supply chain are much greater when the supplier-customer relationship is closer, or when the industry concentration of event firms is higher. Furthermore, the documented spillover effects for suppliers are amplified when the rating change is across two grades (speculative grade ratings and investment grade ratings) or rating agencies have global partners.

Finally, our paper is related to the literature on the importance of investor attention for information diffusion. Investors are more selective in information processing due to limited attention, which can have an impact on the information diffusion across economically linked assets (Peng and Xiong 2006). Hong et al. (2007) show that the slow information diffusion of industry news may open up opportunities for stock return prediction. By contrast, if investor attention is high, information diffusion is likely to be faster. A number of studies have shown that rapid information diffusion in the same investment habitats leads to returns comovement for stocks associated with the habitat. These include habitats such as the S&P 500 (Vijh 1994; Barberis et al. 2005), geography (Pirinsky and Wang 2006; Kumar et al. 2013), and volatility (Dorn and Huberman 2010). More recently, Leung et al. (2017) show that stocks within the same

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cosearch habitats receive high attention from investors and such habitats exhibit high similarity in stock returns or comovement. Similar behavior applies to supply chain stocks. If investors pay attention to the partner firms, then the information diffusion is likely to be high, and, as a result, the stocks are more likely to comove. Our paper contributes to this literature by highlighting that customer-supplier networks with investor attention have valuable asset pricing implications for the spillover effects across debt and equity holders. Our evidence shows that the spillover effects of credit shocks in the supply chain are more pronounced as investor attention increases, suggesting that information about customer (supplier) credit risk should be incorporated into supplier (customer) stock prices when supply chain information improves.

Methodologically, several studies investigate the spillover effects of new information within an industry, across industries, and through supply chains using event-study methodology (e.g., Lang and Stulz 1992; Slovin et al. 1999; Hertzel et al. 2008; Chang et al. 2015; Agca et al. 2022). Our paper contributes to this literature by focusing on the market reaction of an event firm rather than the event itself. The market reaction captures the surprise component of the rating actions better than the rating actions themselves (Joe and Oh 2018). To the best of our knowledge, no existing study has used this method to estimate the spillover effects of generic credit rating actions as we have done, in the supply chain.

2.2. Hypotheses

Capital market efficiency theory suggests that information regarding a firm's financial prospects should flow freely between the stock and the bond markets. Studies have found that new information revealed in the bond market has significant impacts on firms' stock prices (Holthausen and Leftwich 1986; Dichev and Piotroski 2001), which is especially confirmed by some literatures that find abnormal stock returns

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around credit rating events (e.g., Dichev and Piotroski 2001; Jorion and Zhang 2007). On the other hand, supply chains are a likely channel through which financial shocks spread. Supply chains enable flows of physical products and services downstream toward customers and financial flows upstream toward manufacturers and the producers of raw materials. Alongside these physical and financial flows, there exist informational flows that encompass demand forecasts, production capabilities, and the financial strength of companies.

Information is exchanged not only between supply chain partners, but also among their investors. Stock returns of economically linked supply chain partners are correlated because of related fundamental (Hong et al. 2007) and profits (Menzly and Ozbas 2010). It is then expected that investors should pay attention to all stocks in the supply chain and information diffuses quickly in the market. However, the hypothesis of limited investor attention implies that the information diffuses slowly in the market (Hong et al. 2007) and across economically linked assets (Cohen and Frazzini 2008) due to limited attention and investor specialization. The literature on attention provides consistent evidence that investor attention (or conversely, inattention) to firm-specific information and events is associated with positive (negative) capital market effects (Hirshleifer and Teoh 2003; Corwin and Coughenour 2008; Hirshleifer et al. 2009; Chakrabarty and Moulton 2012). High investor attention across supply chain partners represents a higher level of information diffusion that would lead to return comovement—i.e., returns tend to move together in the same direction—for such stocks in the same time period (Barberis et al. 2005).

Market anticipation theory advises that markets tend to react more strongly to unanticipated firm events than to anticipated firm events because unanticipated firm events contain more original information (Smith 1986). Together with that the market

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reaction captures the surprise component of the rating actions more effectively than the rating actions themselves (Joe and Oh 2018), we expect the market response to the event of a firm to be transferred from an event firm to other firms in the supply chain. In other words, the market evaluation of a firm with regard to an event should be crucial in order to determine the characteristics of spillover effects in the supply chain. Furthermore, if investors pay attention to the partner firms, then the information diffusion is likely to be high and as a result, the stocks are more likely to comove (Agarwal et al. 2017). Collectively, these lead to our main hypothesis:

Hypothesis 1a. *Supply chain partners experience spillover effects driven by the market reactions of event firms* (*to credit rating actions*).

Hypothesis 1b. *The spillover effects are enhanced through information diffusion channels as supply chain partners receive more investor attention.*

3. Data and Methodology

3.1. Data

Our data are collected from two sources: WIND database provided by the Wind Information Co., Ltd. for Chinese firms' credit rating action data and scattered customer-supplier relationship data, and China Stock Market & Accounting Research (CSMAR) database for financial and accounting data. In what follows we describe in detail our sample selection procedure.

The primary dataset we examine is the announcement of credit rating actions. Initially, we download 6,555 rating action events from the WIND database between March 2007 and May 2020. Subsequently, we exclude 4,827 events of firms that do not have stock returns. Additionally, we eliminate 120 events that are sequentially close. To focus on the contagion of credit rating actions, we further exclude other firm-

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specific information such as earning announcements, mergers and acquisitions, and seasoned equity offerings within nineteen business days (-9,9) of the event date of the credit rating actions.[2] This rigorous screening process generates our preliminary sample that consists of 1,596 credit rating action events with 315 events for credit watches actions only and 326 events for outlook actions only.

Our other major data set is the lists of customer-supplier relationships. The data for supplier-customer relationships is scattered but available from firm-specific public files.[3] To gather information regarding the major suppliers and customers of Chinese public firms that have undergone credit rating actions between 2007 and 2020, we follow a series of steps. Firstly, we employ a text-matching program to match the event firm's name to the one of WIND segment 'network' file.[4] The files, collected by WIND Data Services, are sourced from the firm's periodic reports (i.e., quarterly reports and annual reports). They contain valuable information about the firm's upstream and downstream business names, stock tickers (if listed), accounts receivable, accounts payable, sales and reporting dates over time period 2006-2020. It is important to note that the WIND network file of a firm is named after the firm itself, and the list of suppliers and customers of the firm may vary across different years. A practical difficulty is that a small number of WIND network files are named after the abbreviated forms of the firms' names. To address this problem, similar to practice in Fee and Thomas (2004), the program requires all of the Chinese Hanzi characters in the reported network file name to be sequentially presented in the potential match. Subsequently, we visually inspect each match generated by the text-matching program to ensure

[2] We apply these screening criteria to event firms and other partner firms in supply chains. Similar practice can be seen in Joe and Oh (2018).
[3] Since 2005, more companies in China have voluntarily disclosed the sales and names of their top five customers and suppliers in their quarterly reports or annual reports.
[4] The text-matching program code is available upon request.

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accuracy. This visual inspection involves some discretion, but it is conducted in a conservative manner that ensures all matches are certain. This matching process produces accurate matches approximately accounting for 94% of total potential matches, excluding all of the possible event–network pairs if there are multiple potential matches and we cannot choose the unique match by screening the available public information (firm web sites, annual reports, and Baidu).

Similarly to Hertzel et al. (2008) who document a negative valuation effect of a customer's bankruptcy filings on its suppliers, all matches are restricted to the two years prior to (and including) the rating action event year. If the matching process produces multiple matches between firm-pairs over the two-year pre-event period, we choose the match-year closest to the event year. Then, we eliminate the cases where the supplier or customer's name is not matched to a unique firm. Subsequently, we exclude the firm-pairs without daily financial data available for the supplier or customer. Finally, to focus on the spillover effects stemming from credit rating action announcements within the supply chain, each event firm must have at least one supplier or customer.

Our sample selection procedure produces a total of 769 events, and the distribution of these events over time is shown in Table 1. The sample contains 276 events of firms that have at least one customer, 624 events firms that have at least one supplier, and 131 events of firms that have at least one customer and one supplier. In addition, our final sample consists of 1,477 event-customer or -supplier observations, spanning across 150 industries. The sample events are concentrated in the time period 2010 to 2020, which contains 711 of the 769 events (92%) of the total sample. The time distribution of our sample of events is representative of total events over this period (as reported in the second column of the table), which breaks down over the same time

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period as 94%. The time-clustering of our sample could reflect variation in firms voluntarily reporting the identity of major customers or suppliers.

For the full sample of 769 events of firms with at least one supplier or customer, we identify a total of 345 suppliers and 139 customers. However, our analysis of supply-chain spillover effects is mainly based on the portfolios of suppliers and customers for each event. The majority of the customer and supplier portfolios have only one firm. As reported in the table, the average numbers of suppliers and customers per event are 1.1 and 1.2.

Motivated by the evidence in prior literature that being a state-owned firm might differentiate a non-state-owned firm in terms of creditworthiness, and its credit rating actions have different influences on financial markets (see, for example, Allen et al. 2005; Cull and Xu 2005; Lu et al. 2012), we divide our event sample into two subsamples: the announcements of state-owned firms and non-state-owned firms. There are 380 (389) events of (non-) state-owned firms that have at least one supplier or one customer in our final sample. For brevity, the distribution of these two subsamples is not tabulated.

Table 2 summarizes the variables for our sample of 769 event firms.[5] The mean (median) book value of total assets is ¥135.1 (15.5) billion. The return on assets (*ROA*), which is calculated by the ratio of net income to total assets, is on average 1.4%. However, the distribution of *ROA* is highly skewed, with a median of 0.9%. The mean (median) leverage is 63.2% (62.0%), and the book to market ratio is 0.81 (0.85). The mean (median) of cash holdings to total assets is 8.5% (6.4%).

[5] Table A1 in Appendix A provides the descriptive statistics of variables for event firms in more details, including the minimum, first quartile, median, mean, third quartile, maximum, and standard deviation.

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Table 2 also shows the descriptive statistics of our event firms in the two subsamples: state-owned firms and non-state-owned firms. The mean *Size* of state-owned firms is ¥22.6 billion, which is larger than that of non-state-owned firms (¥15.7 billion). Particularly, the mean book to market ratio of state-owned firms is significantly higher than that of non-state-owned firms. This is consistent with the idea that state-owned firms tend to be more profitable and mature than non-state-owned firms.

3.2. Methodology

This paper investigates the spillover effects within the supply chain by focusing on the market reactions of event firms to credit rating actions in the supply chain. For each event, we construct a portfolio of suppliers and customers if exist, resulting in a total of 769 portfolios.

Next, we employ the standard event study methodology (Brown and Warner 1985) that calculates the abnormal returns (ARs) and the cumulative abnormal returns (CARs) for event firms by using the market model that follows the event dates of credit rating actions. The model also estimates the ARs and CARs for the supplier and customer portfolios for each event by using the portfolio approach.[6] The AR for the stock of firm *i* at time *t* is computed as follows:

$$AR_{it} = R_{it} - \left(\hat{\alpha}_i + \hat{\beta}_i R_{mt}\right), \tag{1}$$

where R_{it} and R_{mt} are the daily returns for the stock of firm *i* at time *t* and the daily market index (Shanghai Composite Stock Price Index) return at time *t*, respectively. The parameters $\hat{\alpha}_i$ and $\hat{\beta}_i$ are the ordinary least squares estimates of the intercept and slope, respectively, of the market model regression estimated using 180 daily returns

[6] The portfolio approach allows us to diversify away the cross-sectional dependence of individual returns (see, for example, Hertzel et al. 2008; Bae et al. 2008; Joe and Oh 2018).

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beginning with day $t = -200$ and ending with $t = -21$ relative to the credit rating action date. We construct the CAR_i between any two dates t_1 and t_2 by cumulating daily abnormal returns from day t_1 to t_2. We follow a similar approach to Hertzel et al. (2008) to estimate the CARs of supplier and customer portfolios by using equal weighting in order to understand effectively the spillover effects within supply chains.

4. Main results

4.1. Baseline results

Table 3 shows the spillover effects that are driven by the market reactions of event firms. For each event, we form an equal-weighted portfolio of nonevent firms (suppliers and customers) in the supply chain, and calculate the portfolio return. The average return of nonevent firms is the average of these portfolio returns over all events. To unambiguously investigate the relation between the returns of event and nonevent firms, we report the average portfolio returns of nonevent firms separately for two subsamples. These subsamples are formed based on the sign of the CARs of event firms.

The results show that the average two-day and five-day CARs of event firms with negative market reactions to credit rating actions are −4.02% and −4.55%, respectively, both of which are significant at the 1% level. The average two-day and five-day CARs for the corresponding nonevent firms show significant average excess returns of −1.64% and −1.58%, respectively. Meanwhile, we also find that the average two-day and five-day CARs of event firms with positive market reactions to credit rating actions are 3.10% and 3.75%, respectively. Both of those are statistically significant at the 1% level. The average two-day and five-day CARs for the corresponding nonevent firms are 1.25% and 1.64%, respectively, both of which are significant at the 1% level. These findings

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suggest that the credit rating actions of one firm have spillover effects on the other members in its supply chain.

As a further test for spillover effects along supply chains, we conduct regression analyses. In Table 4, we show the market reaction of an event firm to be an important factor in explaining spillover effects within supply chains. We use the five-day CARs(0,4) of nonevent firms (suppliers or customers) in supply chains as the dependent variables. Our key explanatory variable of interest is *Market Reaction*, which is the five-day CAR(0,4) of an event firm. In Panel A, we use *Market Reaction* as the explanatory variable in regression (1). In regression (2), we add firm-level explanatory variables that are listed in Table 2 together with *Credit Rating*, which represents the credit rating of an event firm.[7] In regression (3), we further add year dummies and industry dummies to consider the year and industry fixed effects. Unreported results show that the variance inflation factor (VIF) test has a maximum variance inflation factor of 4.2 and a mean variance inflation factor of 1.2 (omitted for brevity), suggesting that multicollinearity is not a concern.

We focus on the five-day CAR(0,4) rather than the CARs of the short event periods such as the two-day CAR(0,1) in our regression analyses because the latter may take time to be reflected in the market by an event. This is consistent with the conventional wisdom that emerging markets, including the Chinese market, are less efficient than the developed markets. For example, the daily price upper and lower limits for all stocks in the Chinese stock market are 10% and −10%, respectively. This is one example of a hindrance to market efficiency. The price limit rules suggest that market participants may continue to react even after the announcements of credit rating actions. In addition,

[7] Table A1 in Appendix A provides the correlation matrix between all the variables in our main analysis. Table B1 in Appendix B describes categories for credit ratings and the numerical scale used in our analyses.

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we use an ex post event window such as the five-day CAR(0,4), rather than an ex ante event window such as the nine-day CAR(-4,4), to facilitate the analysis of spillover effects driven by the market reactions of event firms. Similarly to Joe and Oh (2018) who report the stock-price effects of Korean business groups by using the announcement of credit rating changes, we focus on the market reactions of event firms to the announcements of credit rating actions rather than the events themselves. Generally, ex ante periods are used on the assumption that information about an event can be leaked before the announcement. Therefore, using the five-day CAR(0,4) is the adaptive way to examine the spillover effects that are driven by market reactions.[8]

The results documented in Panel A in Table 4 show that the initial coefficient on *Market Reaction* in regression (1) is 0.224 and significant at the 1% level. After controlling other firm characteristics in regression (2), and year and industry effects in regression (3), the coefficients on *Market Reaction* remain significantly positive. Evaluating the estimated coefficient in regression (3) at the mean indicates that all else being constant, a 10% increase in the market reaction of an event firm results in about a 1.97% increase in the CARs of other firms in the supply chain. To put it in a different perspective, the abnormal return for nonevent firms increases by 1.65% as the market reaction of an event firm increases by one standard deviation (8.40%). Therefore, the effect of credit rating action events on the market value of nonevent firms seems to be both statistically and economically significant.

The results of Panel B in Table 4 continue to show that spillover effects are pervasive in the subsamples of credit rating changes, credit watches and outlook events. In particular, for rating outlook, the coefficient on *Market Reaction* is significant at the

[8] We also use event windows (0,2), (0,6), (-1,1) and (-2,0) for analysis later to check the robustness of our results. We find that our main conclusions remain intact.

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1% level, with an estimate of 0.876 larger than the estimate for the rating watch (0.543) and rating changes (0.166) scenarios. These results are important because they provide direct evidence that just like the credit rating changes mentioned in the previous literature, credit watch and outlooks events of one firm will also have spillover effects on other firms in the supply chain. It is worth noting that not only credit rating change events, but also credit watch and outlook events often occur. Therefore, in a broader sense, our results clearly show that credit rating action events have supply chain spillover effects. More importantly, our results are consistent with the view that the market evaluation of an event firm plays an important role in determining the characteristics of spillover effects in supply chains.

4.2. Endogeneity tests

Although we report a significant spillover effects in the supply chain, it is possible that our main results are influenced by endogeneity. For example, endogeneity could arise if the market reactions of both event firms and nonevent firms are influenced by unobservable or omitted factors. Another concern is reverse causality. In fact, we investigate the lead and lag relationship between event firms and nonevent firms in supply chains to understand the spillover effects more accurately. Specifically, we regress the lead terms of CARs for nonevent firms on the lag terms of CARs for event firms in supply chains and vice versa. We find that causality dose not run from nonevent firms to event firms (not tabulated for brevity). This practice partly mitigates the issue of reverse causality. Importantly, given that credit rating events are largely exogenous, it is implausible that nonevent firms could accurately predict them, and cause the market reaction of event firms. Nevertheless, we address potential endogeneity concerns in our study in several additional ways, such as using Heckman two-stage regression analysis, using instrumental variable (IV) regression analysis, and

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conducting placebo analysis.

4.2.1. Heckman selection

Companies in China voluntarily disclosed the sales and names of their top five customers and suppliers in their quarterly reports or annual reports since 2005. Because customer and supplier information sometimes is proprietary for a firm, the firm may not willing to disclose it. This may open up the possibility of sample bias. To control for the unobservable omitted variables related to the selection issue, we adopt a Heckman two-stage model.

In the first stage, we model the probability that a firm will disclose information on its top five customers and suppliers using probit regression. The dependent variable is a dummy variable (*Disclosure*) set equal to one if a firm discloses the identities of its major customers and suppliers, and zero otherwise. To satisfy the exclusion restriction, we include the fraction of other companies in the same industry that disclose information on their top five customers and suppliers in the same year (denoted as *Otherratio*) as an independent variable. In an increasingly high-quality disclosure market environment, when the number of other companies in the same industry disclosing the customer and supplier information increases, we expect a company to tend to actively disclose information about its major customers and suppliers, but we have no reason to expect that *Otherratio* will impact its market reactions. Therefore, *Otherratio* can serve as an exogenous determinant of the likelihood that a company might disclose its top five customers and suppliers.

Additionally, we include controls for firm characteristics that may affect the company's disclosure of customer and supplier information, which include return on assets (*ROA*), growth rate (*Growth*), the ratio of total debt to the book value of total assets (*Leverage*), total assets (*Size*), and the number of years elapse since the year of

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the company's IPO (*Firmage*) (Ellis et al. 2012). From the first stage model, we estimate the inverse Mills ratio, which we include in the second stage model to control for potential selection bias.

Panel A of Table 5 reports the results from the first stage probit regression in our Heckman two-stage selection model. Consistent with our expectations that the more other firms in the industry that already disclose customer and supplier information, the more likely a firm is to voluntarily disclose customer and supplier information, we find that the coefficient on the exclusion restriction variable, *Otherratio*, is positive and significant. We also find that *Disclosure* is positively associated with *ROA*, *Growth*, *Size* and *Firmage*, and is negatively correlated with *Leverage*.

Panel B of Table 5 reports the results of controlling for the potential sample selection bias. For brevity, we report only the parameter estimate of *Market Reaction*. We find that the coefficient on *Market Reaction* is significantly positive. This result shows the robustness of our main findings to this selection bias adjustment.

4.2.2. IV regression estimation using heteroskedasticity-based instruments

To further address causality, we employ an IV approach with heteroskedasticity-based instruments following Lewbel (2012). This technique does not rely on an external instrument and instead uses heteroskedasticity in the data to generate internal instruments for the two-stage least squares (2SLS) regression. Lewbel (2012) suggests that this identification technique is especially helpful when external instruments are weak or not available. Furthermore, as discussed in Lewbel (2012), this approach is suitable for many models with error correlation due to unobserved common shocks. Recent studies apply this identification approach to address endogeneity problems (Caliendo et al. 2017; Chen et al. 2021; Hasan et al. 2021).

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In our setting, instruments can be formed from the auxiliary equation's residuals multiplied by each of the included exogenous variables in mean-centered form. Specifically, we follow Lewbel (2012) to generate IVs based on the following equation:

$$IV_j = \left(X_j - \bar{X}\right) \times \varepsilon, \tag{2}$$

where X is each of the exogenous variables in our main model, i.e., *Size*, *ROA*, *Leverage*, *B/M*, *Cash*, and *Credit Rating*; $\bar{X}$ is the mean of the exogenous variable; and ε is the residual from the auxiliary regression of *Market Reaction* (i.e., the event firm's five-day CAR) on all exogenous variables (*Size*, *ROA*, etc.). Thus, we generate as many IVs as the number of exogenous variables in our main model. Next, we use these generated IVs to run the 2SLS regressions based on our main models with *Market Reaction* as the endogenous variable.

The results of Lewbel's analysis are reported in Table 6. The Kleibergen-Paap F statistics indicate that our instruments are sufficiently strong (with p-value less than 0.1). Based on the Hansen J statistics, the overidentification restrictions are satisfied (with p-value above 0.1). Moreover, the results with regard to Shea's partial R^2, F-test of excluded instruments, and Kleibergen-Paap rank LM statistic also show that our choice of the instruments is reasonable. We observe that our main findings remain robust. The coefficient for *Market Reaction* is positive and significant at the 1% level. Thus, Lewbel's method estimation supports our argument that our documented spillover effects propagate through the supply chain and that our results are not an artifact of unobserved common shocks.

4.2.3. Placebo tests

Finally, we conduct three placebo tests to mitigate the potential endogenous concern about omitted variables. The first test is cross-sectional randomization, where the market reactions for event firms are randomized within a given year. The second

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test implements a randomization procedure within a given industry, in which the market reactions for event firms are randomized within a given industry. The third test implements a randomization procedure within a given supplier/customer industry, in which on each event day we randomly select one other firm in the same industry as supplier/customer and calculate its market reaction.

For the first two procedures, we obtain a fictitious market reaction of event firms, defined as *Pseudo Market Reaction*. For the third procedure, we get a fictitious market reaction of other firms in the same industry as supplier/customer, defined as *Pseudo Market Reaction*$_{other}$. These random procedures are all conducted one thousand times to produce one thousand random samples. We then reestimate our main results with the *Pseudo Market Reaction* or *Pseudo Market Reaction*$_{other}$ from these random samples. Specifically, for the first two types of random samples, the market reaction of nonevent firms is regressed on *Pseudo Market Reaction*; while for the last type of random sample, *Pseudo Market Reaction*$_{other}$ is regressed on the market reaction of event firms (i.e., *Market Reaction*). Table 7 summarizes the coefficient estimates. The mean values of estimated coefficients are close to zero in both case with and without control variables. The *t*-tests for the mean show that the average coefficient estimates are not significantly different from zero. The evidence suggests that our main findings are unlikely to be driven by chance.

4.3. Channel analysis

The preceding analysis shows a robust supply chain spillover effect driven by event firm's market reactions. A remaining question is how the spillover effects propagate along the supply chain. To answer this question, we examine a plausible channel through which the market reactions of event firms influence the market reactions of nonevent firms in the supply chain.

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Due to related fundamentals and profits, stock returns of economically linked firms such as supply chain partners are correlated (Hong et al. 2007; Menzly and Ozbas 2010). Investors thus should pay attention to all stocks in the supply chain and information diffuses quickly in the market. However, because of limited attention and investor specialization, the information may diffuse slowly in the market (Hong et al. 2007) and across economically linked assets (Cohen and Frazzini 2008; Menzly and Ozbas 2010). High investor attention across supply chain partners can represent a higher level of information diffusion that would lead to return comovement in the same time period (Barberis et al. 2005). Motivated by these facts, we study whether investor attention to supply chain partners affects the propagation of the supply chain spillovers we document.

We use the number of analysts and the number of media coverage as proxies for investor attention (Hirshleifer and Teoh 2003; Barber and Odean 2008). For analysts, we use information about equity analysts as reported in CSMAR data set, and obtain the total number of analysts following the supplier firm, the customer firm, or both simultaneously in the six months prior to the date of the rating change, respectively. We then define three indicator variables for this analysis: *Supplier Analyst*, *Customer Analyst*, and *Common Analyst*, which equal one if the number of analysts following the supplier firm, the customer firm, or both is higher than its sample median, respectively. For media coverage, we use raw media coverage data from Datago,[9] based on which we obtain the total number of media coverage for supplier firm, customer firm or both in the six months prior to the date of the rating change, respectively. We then define three indicator variables: *Supplier Media*, *Customer Media*, and *Common Media*, which equal one if the number of media coverage for supplier firm, customer firm or both is

[9] See https://www.datago.com.hk.

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higher than its sample median, respectively. We have these variables interact with *Market Reaction*. The results are in Table 8.

Table 8 shows that the propagation of documented spillover effects through supply chains is stronger when supply chain partners are followed by more of the same analysts or reported by more of the same media. For example, in column (4) of Panel A in Table 8, although a 10% increase in the market reaction of an event firm results in about a 1.86% increase in the CARs of other firms in the supply chain, the increase is 4.45% larger (making the overall response 3.4 times larger) if the same analysts are following both supply chain partners. Similarly, as shown in column (4) of Panel B in Table 8, although a 10% increase in the market reaction of an event firm results in about a 2.84% increase in the CARs of other firms in the supply chain, the increase is 1.24% larger (making the overall response 1.4 times larger) if the same media reports both supply chain partners.

To summarize, the amplified reaction when supply chain partners are followed by more of the same analysts or reported by more of the same media points toward the informational channel of shock propagation in the stock market through supply chains. This indicates that our results reflect the propagation of firm level shocks through the information diffusion as investor pay more attention across supply chain partners.

4.4. Cross-sectional tests

4.4.1. Negative spillovers vs. positive spillovers

Our evidence thus far suggests the spillover effects of credit rating actions of one firm on other firms in the supply chain. However, our results are suggestive that the sign of CARs of event firms may leave behind valuable credit-relevant information. Therefore, we ask whether the negative spillovers caused by negative market reactions are stronger than positive spillovers caused by positive market reactions.

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To test this issue, in regression (1) in Table 9 we include the dummy variable *Positive Reaction*, which takes a value of one if the market reaction of an event firm is positive and zero otherwise. We also add the interaction term between *Market Reaction* and *Positive Reaction*, in which *Market Reaction* is the market reaction of an event firm. We find that the coefficients on both *Market Reaction* and its interaction with *Positive Reaction* are negative and significant. This finding suggests that the negative spillovers driven by negative market reactions are stronger than the positive spillovers driven by positive market reactions. Our evidence is consistent with the idea that bad news tends to spread more widely and strongly than good news (Gande and Parsley 2005; Soroka 2006; Ferreira and Gama 2007).

4.4.2. State-owned firms vs. non-state-owned firms

This section investigates whether the spillover effects are different between event firms with state ownership and those without state ownership. To conduct this analysis, we augment the base model by incorporating the dummy variable *State-owned* that takes a value of one if the event firm is state-owned and zero otherwise, and an interaction term between *State-owned* and *Market Reaction*.

In regression (2) in Table 9, the coefficient for the interaction term is negative and significant negative at the 10% level. This result suggests that the positive relationship between the excess returns of event firms and the excess returns of nonevent firms in supply chains is stronger when the event firms are non-state-owned. One potential explanation for these results is that being a non-state-owned firm might differentiate a state-owned firm in terms of creditworthiness (Cull and Xu 2005; Allen et al. 2005), and non-state-owned firms have less availability of credit when monetary policy is tight (Lu et al. 2012). Thus, the state ownership for event firms is relevant information beyond the credit rating level for the market's assessment of its credit-risk, and these

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differences mentioned above lead to greater spillover effects when event firms are non-state-owned.

It is worth noting that the coefficient on *Market Reaction* can be purely interpreted as spillover effects associated with the credit rating actions of non-state-owned event firms. The coefficients estimated on *Market Reaction* in regressions (2) retains the statistical significance at the 1% level. Taken together with the significant coefficient on the interaction term, our evidence shows that the spillover effects are more economically significant when the event firms are non-state-owned. Moreover, the substantially increased adjusted R^2 in regression (2) in Table 9 compared with those in Panel A in Table 4 also reflects that the state ownership status of event firms is a potentially important factor in spillover effects driven by market reactions.

4.4.3. Spillover effects and industry concentration

When the number of suppliers in the market is small or the number of alternative customers is large, suppliers are less likely to be influenced by their major customers (Porter 1979; Cunat 2007). Conversely, when the number of suppliers in the market is small or the number of alternative customers is large, customers are more likely to be influenced by their major suppliers. These arguments suggest that the nonevent firms tend to experience a stronger spillover effect when the industry concentration degree of event firms is higher.

To test this prediction, we use the Herfindahl ratio constructed from the sales of event firms for the whole industry. Using the Herfindahl ratio, we define a dummy variable (*High Concentration*), which takes a value of one if the Herfindahl ratio is above its cross-sectional median for the same year and zero otherwise. The traditional view mention above predicts that the coefficient on the interaction term between *Market Reaction* and *High Concentration* is positive and significant.

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The result is shown in regression (3) in Table 9. The coefficients on *Market Reaction* and the interaction term are both significantly positive. This confirms that the positive relationship between the excess returns for the event firms and the excess returns for the other firms in the supply chain is stronger when the industry concentration degree of event firms is higher.

5. Further analysis

5.1. Suppliers' reactions vs. customers' reactions

The aforementioned analysis has focused mostly on the spillover effects from event firms to the whole nonevent firms. However, the reactions of suppliers and customers can be different since rating changes can affect the bargaining powers between suppliers and customers in different ways. Moreover, the direction of spillovers to nonevent firms along the supply chain can be different when industry peers of event firms react negatively or positively in their stock markets. With these issues in mind, we further look into the average five-day CARs for the event firms, suppliers and customers in supply chains when the industry peers of event firms react negatively or positively in their stock markets. We classify our event samples into two subsamples according to the sign of the CARs of event firms. We then further divide the subsamples according to the sign of the CARs of event firm's industry peers.[10]

Table 10 shows that when peers' reactions are negative and positive, the average five-day CARs of event firms with negative market reactions to credit rating actions are −5.39% and −1.31%, respectively, both of which are significant at the 1% level. When the peers' reactions are negative (positive), the average five-day CARs for the

[10] To see more clearly the difference between the impact of the stock market reaction of event firms and the impact of the event itself, we have shown the market reactions of event firms, suppliers, customers and peers to rating downgrades and upgrades in Table C1 in Appendix C. For brevity, the impact of watches and outlooks is not reported, but is available upon request.

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suppliers and customers show significant average excess returns of −3.07% (−3.08%) and −1.04% (−1.66%), respectively. In parallel, when peers' reactions are negative and positive, the average five-day CARs of event firms with positive market reactions to credit rating actions are 4.42% and 3.84%, respectively. Both of those are statistically significant at the 1% level. When the peers' reactions are negative (positive), the average five-day CARs for the suppliers and customers show significant average excess returns of 2.20% (1.74%) and 0.90% (0.52%), respectively. Overall, these findings show that the credit rating actions of one firm have spillover effects on both suppliers and customers regardless of the reactions of event firm's industry peers. In fact, unreported results (omitted for brevity) indicate that the average negative and positive reactions of event firm's industry peers are not significant in Panels A and B. This suggests that credit events do not belong to industry problems.

It's worth noting that, based on the magnitude of the reactions, the spillover effects to suppliers appear to be greater than those to customers. One possible explanation is that customers largely have stronger bargaining power and thus are less responsive to shocks from suppliers' credit events. Another possibility is that compared to suppliers, customers may have superior understanding and access to recent information about suppliers' credit changes and make more accurate predictions about upcoming changes in demand (Lee et al. 1997).

Further, we conduct regressions using event firm's peer market reaction (*Peer Reaction*) and its interaction with *Market Reaction* as additional control variables. In Panel A of Table 11, the coefficients on *Market Reaction* for both suppliers and customers are significantly positive. This result reconfirms that the spillover effects due to credit rating actions can spread not only upstream, but downstream as well. This finding bolsters our evidence on individual suppliers and customers and constitutes new

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evidence on the extent of spillover effects of credit rating events. Our finding complements that of Alldredge et al. (2019) who find that supplier firms experience negative abnormal returns during the announcement of customer firms' rating downgrades.

On the other hand, in terms of the magnitude of the coefficients, the spillover effects to suppliers seem to be higher than those to customers, which is consistent with the results in Table 10. It should be noted that the coefficients on *Peer Reaction* and *Peer Reaction* × *Market Reaction* are positive, but not significant. This again confirms that the market reactions of event firm's peers have no impact on the spillover effects we documented. Therefore, credit events are largely a matter for individual firms rather than industries.

5.2. Effects of investment grade change

Prior studies find that the impact of rating changes is most significant for those downgrades from investment grades to speculative grades (Goh and Ederington 1999; Creighton et al. 2007; Kisgen 2009; May 2010). Our documented spillover effects may have been more pronounced when the rating change for event firms is across those two grades and, therefore, may be driving our baseline findings. Thinking about this issue prompts us to wonder if stronger spillovers should be observed when the rating change is across two grades.

To this end, we create a dummy variable *Across* that is equal to one if the rating change is across two grades and zero otherwise. Then we interact this dummy variable with *Market Reaction*. Additionally, we replace credit rating *Credit Rating* (an ordered categorical variable) with *IG*, which is an indicator of one if the rating is investment grade and zero otherwise. This helps separate the firm characteristics into two distinct groups.

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We incorporate these variables into our base model, and report the regression results in Panel B of Table 11. For suppliers, the coefficient on *Across* × *Market Reaction* is significantly positive. Our evidence is in line with the existing literature that financial markets react more strongly to downgrade and upgrade announcements for firms with rating change across two grades (speculative grade ratings and investment grade ratings) (Li et al. 2006; Kisgen 2009; May 2010). For customers, the coefficient is positive, but not significant. This suggests that the investment grade change is not a potentially important influencing factor in our documented spillovers to customers.

5.3. Effects of global rating partners

Our credit rating data are from several rating agencies and some of them have global partners such as S&P, Moody's and Fitch. One may feel that those rating agencies with global famous partners maybe more independent even though they may not have active role in rating assignments but have reputation concerns especially for issuing rating on state-owned firms. Thus, we investigate whether stronger spillovers are observed when rating agencies have global partners.[11]

To test our intuition above, we define a dummy variable *Global*, which takes a value of one if the rating agency has global partners and zero otherwise. Then we interact *Global* with *Market Reaction*. We extend our base model by including these variables, and report the regression results in Panel C of Table 11. For suppliers, the coefficient on *Global* × *Market Reaction* is significantly positive. This suggests that our documented spillover effects for suppliers are amplified when rating agencies have global partners. This finding is consistent with the idea that stock markets react more strongly to announcements of global ratings agencies than to those of local counterparts

[11] Table D1 in Appendix D provides a brief overview of Chinese credit rating agencies.

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(Li et al. 2006). For customers, the coefficient is not significant, suggesting that global rating partners are not an important factor in the spillovers to customers.

5.4. Effects of business relationships in the supply chain

Spillover effects should be more dominant when the business relationship between an event firm and other members (i.e., suppliers and customers) in the supply chain is closer. In what follows we provide additional evidence on this condition. Following common practices in the literature, we measure the extent of a business relationship by *Sales* (i.e., the average sales to supplier firms or customer firms divided by the total sales of an event firm). Using this variable, we construct a dummy variable. It takes a value of one if the variable is above its cross-sectional median and zero otherwise, and is denoted as *Closer*.

The result is shown in Panel C of Table 11. For suppliers, the coefficient on *Market Reaction* is positive and significant at the 1% level. We are particularly interested in the coefficient on the interaction term between *Market Reaction* and *Closer*, which is significantly positive. Meanwhile, for customers these coefficients are positive and significant at the 5% level. These findings suggest that the spillover effects to suppliers and customers are both stronger when the level of business relationship is higher. Our results complement the evidence in Houston et al. (2016) who find that the impact of a firm's bankruptcy on suppliers is greater when there is a stronger supplier-customer relationship.

6. Robustness test

6.1. Impact of overlapping estimation and event periods

A potential concern is the impact of estimation periods on the independence of the computed returns when we estimate the abnormal returns. Since we use the market-

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model approach to estimate abnormal returns for the events and since the estimation periods (day −200 to day −20) of different events overlap in many cases, it is likely that the *t*-statistics in the analyses of the abnormal returns are biased upward. To see whether the overlapping estimation period affects our results in a significant way, we repeat all the analyses described above using the market-adjusted-return model (MA). As displayed in Panel A in Table 12, our main results are qualitatively similar to those reported above. In addition, we experiment with the constant-mean-return model (CM) for which the benchmark return is estimated by averaging the returns from day −244 to day −6, and find that our main conclusions remain intact. Therefore, our results do not seem to be affected by overlapping estimation periods.

6.2. Different event windows

So far we have focused on the event window from day 0 to day +4 to account for the potential effect of the daily price limits in the Chinese stock markets on market reactions. To check the robustness of our results, we use the same explanatory variables as those in Tables mentioned above, but change the definition of the event window from (0,4) to (0,2) and (0,6). In addition, we use the event windows such as (-1,1) and (-2,0) to consider the likelihood that stock markets can react to the credit change of event firms before the rating announcements (Norden and Weber 2004). As displayed in Panel B in Table 12, we obtain qualitatively identical results, suggesting that the event window of (0, 4) used in the previous analyses is robust enough to illustrate the spillover effects.

7. Conclusions

In this paper, we explore the spillover effects of supply chains in China. To examine the existence of spillover effects between suppliers and customers in the

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supply chain, we focus on the market reactions of event firms to announcements of credit rating actions such as rating changes, credit watches, and outlooks. We provide evidence of a positive association between the abnormal returns of event firms and the abnormal returns of supply chain partners (suppliers and customers) around the announcements of event firms' credit rating actions. The spillover effects are substantially stronger for supply chain partners that receive more investor attention. This points to information diffusion as an important channel driving shock propagation. Moreover, we find that negative spillovers caused by negative market reactions are more dominant than positive spillovers caused by positive market reactions. This finding supports the idea that public responses to negative news in general are much larger than those to positive news.

Finally, we find that several supply chain and firm attributes affect the documented spillover effects. In particular, the spillover effects are stronger when event firms are non-state-owned. This is consistent with the fact that market participants are largely more concerned about the credit rating actions of private firms, because they are more dependent on private investments and banks loans for growth. We also find that both closer supplier-customer business relationship and higher industry concentration of event firms amplify the spillover effects. On the other hand, the spillover effects to suppliers rather than to customers are amplified when the rating change of even firms is across two grades (investment grades and speculative grades) or rating agencies have global partners.

To summarize, we show an important channel—supply chains—for credit shock propagation in the stock market for both adverse and favorable credit shocks. Our findings suggest that credit rating actions convey new price-relevant information to stock investors across the supply chain and that investor attention improves information

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diffusion. Our results also provide insights into how the customer-supplier economic linkage plays a role in stock asset pricing during firm credit rating events.

We caveat that our study is not without limitations. In this study, we conclude that supply chain partners experience spillover effects driven by the market reactions of event firms to credit rating actions. We only identify the information diffusion channel that drives our documented spillover effects. In the future, we can extend the research to investigate more thoroughly the underlying economic channels of spillover effects. In addition, with the accumulation of finer data, we can also examine more potential factors that determine the degree of spillover effects, such as loan or tax policy. We believe that our study focusing on the market reaction of event firms to explore the spillover effects along supply chain partners will provide a new perspective and approach to aid future pursuits along these lines by researchers.

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Table 1: Sample description

Year	No. of events	No. with customer or supplier match	No. of supplier portfolio	No. of customer portfolio	No. of event-customer or supplier	No. of industry
2007	13	9	8 (1.1)	6 (1.2)	24	3
2008	33	21	17 (1.1)	8 (1.1)	39	6
2009	48	28	22 (1.1)	11 (1.0)	66	7
2010	65	40	33 (1.1)	16 (1.0)	80	7
2011	61	35	26 (1.0)	18 (1.2)	80	8
2012	90	52	43 (1.1)	17 (1.1)	98	9
2013	117	60	53 (1.1)	18 (1.1)	124	12
2014	137	72	58 (1.1)	28 (1.1)	130	9
2015	182	93	75 (1.1)	33 (1.1)	185	14
2016	213	103	83 (1.1)	36 (1.1)	188	13
2017	186	75	62 (1.2)	25 (1.3)	134	12
2018	162	59	51 (1.3)	19 (1.2)	111	16
2019	167	75	55 (1.3)	28 (1.5)	139	17
2020	122	47	38 (1.1)	13 (1.2)	79	17
Total	1,596	769	624 (1.1)	276 (1.2)	1,477	150

Notes: This table describes the distribution of credit rating change events by year from March 2007 to May 2020. It reports separately the number of all events, the number of events of firms that have at least one supplier or customer around the event day, the number of events of firms with supplier portfolio, the number of events of firms with customer portfolio, the number of event-supplier or -customer observations, and the number of industry coverage in terms of WIND secondary industry classification standard. The numbers in parentheses are average number of suppliers or customers in the portfolio.

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Table 2: Summary statistics of event firms' characteristics

	Full sample	State-owned	Non-state-owned	
Variables	Mean (median)	Mean (median)	Mean (median)	*t*-test
Total Asset	135.1 (15.5)	156.6 (21.1)	108.7 (11.5)	0.382
Size	19.4 (7.9)	22.6 (9.6)	15.7 (6.0)	0.918
ROA	0.014 (0.009)	0.012 (0.010)	0.016 (0.007)	-0.130
Leverage	0.632 (0.620)	0.641 (0.636)	0.620 (0.594)	1.141
B/M	0.808 (0.851)	0.841 (0.869)	0.768 (0.812)	3.655***
Cash	0.085 (0.064)	0.089 (0.074)	0.081 (0.053)	0.879

Notes: This table shows the summary statistics of event firms in our sample. *Total Asset* is the book value of total assets (billion ¥). *Size*, *ROA*, *Leverage*, *B/M*, and *Cash* are calculated by the market value of equity, the ratio of net income to total assets, the ratio of total debt to total assets, the ratio of the book value of equity to the market value of equity, and the ratio of cash to total assets, respectively. *** indicates significance at the 1% level.

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Table 3: Spillover effects along supply chains

	Event firms			Nonevent firms		
Event windows	*N*	CAR	*t*-stat	*N*	CAR	*t*-stat
Panel A: Negative reactions of event firms (negative spillovers)						
Two-day CAR(0,1)	396	-4.02	-6.95***	396	-1.64	-4.32***
Five-day CAR(0,4)	398	-4.55	-7.41***	398	-1.58	-3.51***
Panel B: Positive reactions of event firms (positive spillovers)						
Two-day CAR(0,1)	373	3.10	14.87***	373	1.25	3.38***
Five-day CAR(0,4)	371	3.75	14.53***	371	1.64	4.12***

Notes: This table presents the spillover effects that are driven by the market reactions of event firms. The average two-day and five-day CARs are reported for the event firms and other firms in supply chains. We classify our event samples into two subsamples according to the sign of the CARs of event firms. The subsample in Panel A includes event firms with negative CARs (negative spillovers), and the subsample in Panel B includes event firms with positive CARs (positive spillovers). *** indicates significance at the 1% level.

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Table 4: The role of market reactions in spillovers

Panel A: Full sample

	(1)	(2)	(3)
Market Reaction	0.224***	0.228***	0.197***
	[0.035]	[0.036]	[0.039]
Size		0.002	0.005
		[0.003]	[0.004]
ROA		-0.025	-0.029
		[0.044]	[0.050]
Leverage		0.024	0.022
		[0.016]	[0.019]
B/M		0.027**	0.031*
		[0.014]	[0.016]
Cash		0.008	0.006
		[0.033]	[0.040]
Credit Rating		0.001	0.001
		[0.001]	[0.001]
Constant	-0.001	-0.069	-0.121*
	[0.003]	[0.044]	[0.070]
Year fixed effects	No	No	Yes
Industry fixed effects	No	No	Yes
Adjusted R^2	0.087	0.107	0.167
F-value	41.400	7.067	5.851
Observations	769	758	758

Panel B: Different rating actions

Sample	Rating changes	Watch only	Outlook only
Market Reaction	0.166***	0.543***	0.876***
	[0.043]	[0.162]	[0.184]
Control variables	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Adjusted R^2	0.214	0.362	0.352
F-value	1.695	3.877	5.411
Observations	455	148	155

Notes: This table presents the estimates of OLS regressions for the whole sample and different credit rating actions' subsamples in Panels A and B, respectively. The dependent variable is the five-day CAR for nonevent firms in supply chains. The explanatory variable of interest is *Market Reaction* (i.e., the event firm's five-day CAR). Control variables with regard to an event firm include *Size* (i.e., the market value of equity), *ROA* (i.e., the ratio of net income to total assets), *Leverage* (i.e., the ratio of total debt to total assets), *B/M* (i.e., the ratio of the book value of equity to the market value of equity), *Cash* (i.e., the ratio of cash to total assets), and *Credit Rating* (i.e., the credit rating level). The standard errors reported in parentheses are robust and clustered at the event-firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Table 5: Sample-selection bias

Panel A: First-stage estimation to get inverse Mills ratio

	Disclosure
Otherratio	7.712***
	[0.159]
ROA	0.242**
	[0.098]
Growth	0.160
	[0.190]
Leverage	-0.256***
	[-0.101]
Size	0.032**
	[0.013]
Firmage	0.053**
	[0.022]
Constant	1.471***
	[0.555]
Year fixed effects	Yes
Industry fixed effects	Yes
Max-rescaled R^2	0.350
Observations	23017

Panel B: Regression controlling for selection bias

	Nonevent firms' CAR(0,4)
Market Reaction	0.183**
	[0.094]
Constant	1.408
	[0.922]
Control variables	Yes
Inverse Mills ratio	Yes
Year fixed effects	Yes
Industry fixed effects	Yes
Adjusted R^2	0.173
Observations	754

Notes: This table reports the test results for potential selection bias using Heckman two-stage method. In Panel A, we obtain the inverse Mills ratio from the first-stage probit model of whether a firm discloses customer and supplier information. *Disclosure* is a dummy variable with a value of one if a firm discloses customer/supplier information, and zero otherwise. *Otherratio* is the fraction of industry peers that disclose their customer/supplier information. *ROA*, *Growth*, *Leverage*, *Size*, *Firmage*, *Market Reaction* are respectively the ratio of net income to total assets, the growth rate, the ratio of total debt to total assets, the market value of equity, the firm age, and the five-day CAR. In Panel B, we add the estimated inverse Mills ratio to control for self-selection bias. The explanatory variable of interest is *Market Reaction* (i.e., the event firm's five-day CAR). Control variables with regard to an event firm include *Size* (i.e., the market value of equity), *ROA* (i.e., the ratio of net income to total assets), *Leverage* (i.e., the ratio of total debt to total assets), *B/M* (i.e., the ratio of the book value of equity to the market value of equity), *Cash* (i.e., the ratio of cash to total assets), and *Credit Rating* (i.e., the credit rating level). Robust standard errors clustered at the event-firm level are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Table 6: IV regression using heteroskedasticity-based instruments

	First stage	Second stage
	Market Reaction	Nonevent firms' CAR(0,4)
Market Reaction		0.690***
		[0.263]
Size	-0.003	0.001
	[0.003]	[0.003]
ROA	0.019	0.008
	[0.049]	[0.037]
Leverage	-0.022	0.030
	[0.020]	[0.026]
B/M	0.023	0.023**
	[0.016]	[0.012]
Cash	0.034	0.008
	[0.039]	[0.037]
Credit Rating	-0.001	0.001
	[0.013]	[0.002]
IV_{Size}	-0.231***	
	[0.073]	
IV_{ROA}	3.214**	
	[1.256]	
$IV_{Leverage}$	-0.345	
	[0.362]	
$IV_{B/M}$	-1.409***	
	[0.223]	
IV_{Cash}	-5.376***	
	[0.786]	
$IV_{Credit\ Rating}$	0.006	
	[0.022]	
Constant	0.052	-0.034
	[0.050]	[0.064]
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Observations	758	758
Shea's partial R^2	0.462	
F-test of excluded instruments	5.378	
p-value	0.01	
Kleibergen-Paap rank LM-statistic	13.249	
p-value	0.03	
Kleibergen-Paap *F*-statistic	17.3	
p-value	0.01	
Hansen *J* statistic	8.88	
p value	0.11	

Notes: This table reports the result of two-stage least squares (2SLS) instrumental variable regressions. We follow Lewbel (2012) in generating heteroskedasticity-based instruments, which uses $\left(X-\bar{X}\right)\times\varepsilon$ as instruments, where X is each of the exogenous variables in our main model. The explanatory variable of interest is *Market Reaction*. Control variables with regard to an event firm include *Size* (i.e., the market value of equity), *ROA* (i.e., the ratio of net income to total assets), *Leverage* (i.e., the ratio of total debt to total assets), *B/M* (i.e., the ratio of the book value of equity

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to the market value of equity), *Cash* (i.e., the ratio of cash to total assets), and *Credit Rating* (i.e., the credit rating level). Robust standard errors clustered at the event-firm level are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Table 7: Placebo tests

Panel A: Random market reaction within year

	Mean	*t*-stat	*p*-value
Pseudo Market Reaction	0.001	1.040	0.297
Pseudo Market Reaction with controls	0.001	0.980	0.328

Panel B: Random market reaction within industry

	Mean	*t*-stat	*p*-value
Pseudo Market Reaction	-0.000	-0.410	0.682
Pseudo Market Reaction with controls	-0.001	-0.540	0.592

Panel C: Random market reaction of other firms within a given supplier/customer-industry

	Mean	*t*-stat	*p*-value
Pseudo Market Reaction$_{other}$	0.001	0.390	0.694
Pseudo Market Reaction$_{other}$ with controls	0.000	0.142	0.886

Notes: This table reports the results from placebo tests. In Panels A and B, *Pseudo Market Reaction* is generated by randomizing the market reactions of event firms within a given year and industry, respectively. In Panel C, *Pseudo Market Reaction*$_{other}$ is generated by implementing a randomization procedure within a given supplier/customer industry, in which on each event day we randomly select one other firm in the same industry as supplier/customer and calculate its market reaction. The average value of coefficient estimates on the *Pseudo Market Reaction* (in Panels A and B) or *Market Reaction* (in Panel C) from 1000 trials, together with *t*-statistics and *p*-values from the *t*-test for the mean are presented. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Table 8: The information diffusion channel

Panel A: Analyst

	(1)	(2)	(3)	(4)
Market Reaction	0.194***	0.194***	0.176***	0.186***
	[0.041]	[0.040]	[0.039]	[0.042]
Supplier Analyst	-0.010			-0.009
	[0.012]			[0.006]
Market Reaction × *Supplier Analyst*	0.087			0.007
	[0.120]			[0.125]
Customer Analyst		0.010		0.013
		[0.009]		[0.009]
Market Reaction × *Customer Analyst*		0.004		-0.004
		[0.010]		[0.010]
Common Analyst			-0.001	-0.001
			[0.008]	[0.009]
Market Reaction × *Common Analyst*			0.461***	0.445***
			[0.160]	[0.169]
Constant	-0.149**	-0.095	-0.136*	-0.134*
	[0.071]	[0.071]	[0.071]	[0.073]
Control variables	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.274	0.280	0.328	0.365
Observations	758	758	758	758

Panel B: Media

	(1)	(2)	(3)	(4)
Market Reaction	0.316***	0.309***	0.280***	0.284***
	[0.060]	[0.064]	[0.061]	[0.065]
Supplier Media	0.002			0.001
	[0.005]			[0.006]
Market Reaction × *Supplier Media*	-0.044			-0.107
	[0.140]			[0.143]
Customer Media		0.003		0.004
		[0.009]		[0.009]
Market Reaction × *Customer Media*		0.015		-0.051
		[0.152]		[0.154]
Common Media			0.001	0.001
			[0.002]	[0.002]
Market Reaction × *Common Media*			0.106***	0.124**
			[0.005]	[0.059]
Constant	0.197**	0.194**	0.176*	0.185*
	[0.097]	[0.098]	[0.097]	[0.100]
Control variables	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.293	0.298	0.324	0.358
Observations	758	758	758	758

Notes: This table reports the estimates of OLS regressions for the effect of investor attention on the spillover effects that are driven by the market reactions of event firms. The dependent variable is the CAR(0,4) for nonevent firms in supply chains. *Market Reaction* is the CAR(0,4) for event firms. *Supplier* △, *Customer* △ and *Common* △ are three dummy variables (△=*Analyst/Media*) with a value of 1 if a target channel proxy variable is higher than median value, and 0 otherwise. Control variables with regard to an event firm include *Size* (i.e., the market value of equity), *ROA* (i.e., the ratio of net income to total assets), *Leverage* (i.e., the ratio of total debt to total assets), *B/M* (i.e., the ratio of the book value of equity to the market value of equity), *Cash* (i.e., the ratio of cash to total assets), and *Credit Rating* (i.e., the credit rating level). Robust standard errors clustered at the

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event-firm level are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Table 9: Cross-sectional tests

	(1)	(2)	(3)
	Asymmetric spillovers	State ownership	Industry concentration
Market Reaction : *a*	0.405***	0.222***	0.408***
	[0.121]	[0.053]	[0.094]
b	-0.013**	-0.006	0.001
	[0.006]	[0.007]	[0.006]
$a \times b$	-0.306**	-0.125*	0.242**
	[0.133]	[0.064]	[0.101]
Control variables	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Adjusted R^2	0.258	0.249	0.154
Observations	758	758	758

Notes: This table reports the estimates of OLS regressions for cross-sectional tests. The dependent variable is the CAR(0,4) for nonevent firms in supply chains. $a \times b$ is the explanatory variable of interest, defined as the interaction between *Market Reaction* (*a*) and cross-sectional variable (*b*), where *b* represents the following dummy variables: *Positive Reaction*, *State-owned*, or *High Concentration*. *Positive Reaction* is a dummy variable with a value of one if the market reaction of an event firm is positive and zero otherwise. *State-owned* is an indicator variable with a value of one if the event firm is state-owned and zero otherwise. *High Concentration* takes a value of one if the Herfindahl ratio is above its cross-sectional median for the same year and zero otherwise. *Market Reaction* is the CAR(0,4) for event firms. Control variables with regard to an event firm include *Size* (i.e., the market value of equity), *ROA* (i.e., the ratio of net income to total assets), *Leverage* (i.e., the ratio of total debt to total assets), *B/M* (i.e., the ratio of the book value of equity to the market value of equity), *Cash* (i.e., the ratio of cash to total assets), and *Credit Rating* (i.e., the credit rating level). Robust standard errors clustered at the event-firm level are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Table 10: Spillover effects along supply chains: supplier vs. customer

Peers' reactions	Event firms			Suppliers			Customers		
	N	CAR	*t*-stat	*N*	CAR	*t*-stat	*N*	CAR	*t*-stat
Panel A: Negative reactions of event firms (negative spillovers)									
Negative	224	-5.39	-9.72***	224	-3.07	-2.01**	224	-1.04	-1.98**
Positive	174	-1.31	-2.79***	174	-3.08	-2.93***	174	-1.66	-2.23***
Panel B: Positive reactions of event firms (positive spillovers)									
Negative	192	4.42	2.44***	192	2.20	2.86***	192	0.90	2.42***
Positive	179	3.84	8.69***	179	1.74	2.47***	179	0.52	2.18**

Notes: This table presents the spillovers to suppliers and customers. The average five-day CARs are reported for the event firms, suppliers and customers in supply chains when event firm's peers react negatively or positively in their stock markets. We classify our event samples into two subsamples according to the sign of the CARs of event firms. We then further divide the subsamples according to the sign of the CARs of event firm's industry peers. The subsample in Panel A includes event firms with negative CARs (negative spillovers), and the subsample in Panel B includes event firms with positive CARs (positive spillovers). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Table 11: Further analysis: the regression results of spillover effects in the supply chain

Panel A: The impact of event firms' peer market reactions

	Supplier CAR(0,4)	Customer CAR(0,4)
Market Reaction	0.304***	0.270**
	[0.075]	[0.132]
Peer Reaction	0.101	0.068
	[0.162]	[0.084]
Peer Reaction × Market Reaction	0.109	0.132
	[0.146]	[0.338]
Control variables	Yes	Yes
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Adjusted R^2	0.283	0.343
Observations	620	271

Panel B: The impact of investment grade change

	Supplier CAR(0,4)	Customer CAR(0,4)
Market Reaction	0.388***	0.305*
	[0.067]	[0.166]
Across	-0.023	0.011
	[0.024]	[0.019]
Across × Market Reaction	1.104**	0.246
	[0.460]	[0.256]
Control variables	Yes	Yes
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Adjusted R^2	0.209	0.317
Observations	620	271

Panel C: The impact of global rating partners

	Supplier CAR(0,4)	Customer CAR(0,4)
Market Reaction	0.302***	0.254**
	[0.071]	[0.145]
Global	0.003	-0.007
	[0.011]	[0.012]
Global × Market Reaction	0.465**	0.145
	[0.187]	[0.215]
Control variables	Yes	Yes
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Adjusted R^2	0.209	0.305
Observations	620	271

Panel D: The impact of business relationships in the supply chain

	Supplier CAR(0,4)	Customer CAR(0,4)
Market Reaction	0.147***	0.135**
	[0.046]	[0.058]
Closer	-0.006	0.009
	[0.009]	[0.018]
Closer × Market Reaction	0.582***	0.373**
	[0.114]	[0.165]

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Control variables	Yes	Yes
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Adjusted R^2	0.355	0.362
Observations	620	271

Notes: This table reports the estimates of OLS regressions for the further analysis of spillover effects. We consider the impact of event firm's peer market reactions in panel A, investment grade change in Panel B, global rating partners in Panel C, and business relationships in Panel D. *Market Reaction* is the CAR(0,4) for event firms. *Peer Reaction* is the CAR(0,4) for event firm's industry peers. *Across*, *Global* and *Closer* are three dummy variables: the first takes a value of one if the rating change is across two grades and zero otherwise; the second takes a value of one if the rating agency has global partners and zero otherwise; and the last takes a value of one if the event firm's sales to suppliers or customers is above their sample median and zero otherwise. Control variables with regard to an event firm include *Size* (i.e., the market value of equity), *ROA* (i.e., the ratio of net income to total assets), *Leverage* (i.e., the ratio of total debt to total assets), *B/M* (i.e., the ratio of the book value of equity to the market value of equity), *Cash* (i.e., the ratio of cash to total assets), and *Credit Rating* (i.e., the credit rating level). The standard errors reported in parentheses are robust and clustered at the event-firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Table 12: Robustness tests on the role of market reactions in spillovers

Panel A: Results based on different benchmark models

	MA	CM
Market Reaction	0.088**	0.070**
	[0.044]	[0.032]
Control variables	Yes	Yes
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Adjusted R^2	0.115	0.103
Observations	758	758

Panel B: Results based on different event windows

	CAR(0,2)	CAR(0,6)	CAR(-1,1)	CAR(-2,0)
Market Reaction	0.179***	0.210***	0.314***	0.574***
	[0.044]	[0.040]	[0.059]	[0.071]
Control variables	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.159	0.182	0.281	0.379
Observations	758	758	758	758

Notes: In Panel A, the OLS regressions are repeated using the CAR(0,4) from the market-adjusted-return model (MA) and constant-mean-return model (CM). In Panel B, the OLS regressions are repeated using CAR(0,2), CAR(0,6), CAR(-1,1) and CAR(-2,0). The explanatory variable of interest is *Market Reaction* (i.e., the event firm's five-day CAR). Control variables with regard to an event firm include *Size* (i.e., the market value of equity), *ROA* (i.e., the ratio of net income to total assets), *Leverage* (i.e., the ratio of total debt to total assets), *B/M* (i.e., the ratio of the book value of equity to the market value of equity), *Cash* (i.e., the ratio of cash to total assets), and *Credit Rating* (i.e., the credit rating level). Robust standard errors clustered at the event-firm level are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Appendix A: Descriptive statistics and correlation coefficients

Table A1: Descriptive statistics and correlation matrix

Panel A: The descriptive statistics of the variables in the main analysis

Variable	Observations	Mean	Minimum	First quartile	Median	Third quartile	Maximum	Standard deviation
CAR(0, 4)	769	-0.003	-0.512	-0.020	0.001	0.022	0.194	0.060
Market Reaction	769	-0.006	-0.518	-0.021	-0.001	0.020	0.145	0.076
Size	769	19.429	0.313	3.885	7.925	17.349	137.718	74.554
ROA	769	0.014	-0.926	-0.004	0.009	0.011	0.121	0.066
Leverage	769	0.632	0.195	0.521	0.620	0.755	1.380	0.178
B/M	769	0.808	0.236	0.681	0.851	0.952	1.316	0.199
Cash	758	0.085	0.000	0.034	0.064	0.112	0.589	0.081
Credit Rating	769	10.385	1.000	10.000	11.000	12.000	14.000	2.958

Panel B: The Pearson correlation coefficients between all the variables in the main analysis

	CAR(0, 4)	*Market Reaction*	*Size*	*ROA*	*Leverage*	*B/M*	*Cash*	*Credit Rating*
CAR(0, 4)	1							
Market Reaction	0.306***	1						
Size	-0.001	-0.021	1					
ROA	-0.045	-0.034	0.034	1				
Leverage	0.088*	-0.033	0.100**	-0.159***	1			
B/M	0.129**	0.052	0.057	0.062	0.104**	1		
Cash	-0.004	0.063	-0.035	-0.121**	-0.208***	-0.073		
Credit Rating	-0.018	-0.069	0.301***	0.187***	-0.241***	0.139***	0.102**	1

Notes: This table reports the descriptive statistics and Pearson correlation coefficients of the variables used in the main analysis in Panel A and Panel B, respectively. *CAR*(0, 4) is the five-day cumulative abnormal returns for nonevent firms. The variables with regard to an event firm include *Market Reaction* (i.e., the five-day CAR), *Size* (i.e., the market value of equity), *ROA* (i.e., the ratio of net income to total assets), *Leverage* (i.e., the ratio of total debt to total assets), *B/M* (i.e., the ratio of the book value of equity to the market value of equity), *Cash* (i.e., the ratio of cash to total assets), and *Credit Rating* (i.e., the credit rating level). *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Appendix B: The scale of the rating levels for the Chinese main rating agencies

Table B1: Credit rating scale

Rating	Numeric value assigned
AAA	14
AAA-	13
AA+	12
AA	11
AA-	10
A+	9
A	8
A-	7
BBB+	6
BBB	5
BBB-	4
BB+, BB, BB-	3
B+, B, B-	2
CCC, CC, C	1

Notes: This table describes the categories that are used for credit ratings. It also presents the numerical scale used in the paper.

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Appendix C: The market reactions of event firms, suppliers, customers, and peers to rating events

Table C1: The market reactions of event firms, suppliers, customers and peers to credit rating events

	Event firms			Suppliers			Customers			Peers		
	N	CAR	*t*-stat	*N*	CAR	*t*-stat	*N*	CAR	*t*-stat	*N*	CAR	*t*-stat
Panel A: Rating downgrades												
CAR(0,1)												
Negative	77	-7.60	-2.72***	104	-5.00	-3.69***	43	-3.81	-3.56***	95	-0.40	-0.69
Positive	96	3.38	5.83***	122	3.81	3.98***	62	4.69	3.78***	78	0.28	0.91
CAR(0,4)												
Negative	71	-6.80	-3.88***	81	-5.19	-2.97***	50	-5.79	-3.19***	92	-0.48	-0.53
Positive	102	3.69	5.10***	145	3.43	2.80***	55	4.96	2.95***	81	0.28	1.04
CAR(-1,1)												
Negative	86	-4.78	-4.01***	110	-3.40	-3.19***	55	-4.13	-2.53***	90	-0.41	-0.59
Positive	87	2.83	4.91***	116	2.93	2.88***	50	3.980	2.62***	83	0.21	0.86
Panel B: Rating upgrades												
CAR(0,1)												
Negative	162	-3.21	-7.58***	168	-5.17	-3.71***	93	-3.51	-3.55***	144	-0.40	-0.80
Positive	120	2.42	7.24***	226	3.06	6.50***	73	2.27	4.44***	138	0.24	1.26
CAR(0,4)												
Negative	153	-4.23	-5.71***	150	-7.20	-3.44***	86	-4.07	-4.03***	132	-0.60	-0.66
Positive	129	2.91	6.63***	244	3.18	5.48***	80	2.53	3.91***	150	0.33	1.15
CAR(-1,1)												
Negative	170	-6.80	-3.88***	181	-5.19	-2.97***	90	-5.79	-3.19***	143	-0.51	-0.69
Positive	112	3.69	5.10***	213	3.43	2.80***	76	4.96	2.95***	139	0.30	1.09

Notes: This table shows the market reactions of event firms, suppliers, customers and peers to rating downgrades and upgrades in Panel A and Panel B, respectively. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Appendix D: A brief overview of Chinese credit rating agencies

Table D1 provides the nine major domestic credit rating agencies (CARs). They include three independent agencies with full accreditation in both the interbank market and the exchange-based market: Dagong, Brilliance and Jincheng. Brilliance has a technical partnership with Standard & Poor's that it formed in 2008 for the purpose of sharing knowledge and experience. The two domestic independent agencies of FarEast and Pengyuan have full accreditation in the exchange-based market, and restricted accreditation in the interbank market. In addition, there are two domestically owned and capitalized multi-agency groups, Chengxin and Lianhe, which each effectively comprise two entities: a domestically-funded entity having accreditation in the exchange-based market, and a joint-venture entity with minority foreign participation limited to rating in the inter-bank market. Chengxin's joint venture is with Moody's and Lianhe's with Fitch. Ratings from all the above domestic agencies are issuer-paid.

The last five columns of Table D1 list the approvals from the five government regulatory bodies for each CAR.[12] The Securities and Exchange Commission (SEC) is the sole regulator of the CRAs in the US, but multiple Chinese government regulators have their own approved CRAs. Two CARs (Brilliance and Dagong) have approvals from all five regulators. On the other hand, ChengxinI has approvals from only two regulators. As pointed out in Jiang and Packer (2019), though the three main global agencies have either minority shareholder stakes in the joint venture entities or technical partnerships with domestic agencies as mentioned above, the global agencies' themselves do not play an active role in those entities' assignment of ratings during the period under investigation.

Table D1: List of major Chinese credit rating agencies

Name	Global partner	Website	Head-quarter	Regulatory approval				
				PBoC	CIRC	CSRC	NDRC	NAFMII
Brilliance	S&P*	www.shxsj.com	Shanghai	Yes	Yes	Yes	Yes	Yes
ChengxinS	Moody's*	www.ccxi.com.cn	Beijing	Yes	No	Yes	Yes	Yes
Lianhe Fitch	Fitch*	www.lhratings.com	Beijing	Yes	No	Yes	Yes	Yes
Dagong	None	www.dagongcredit.com	Beijing	Yes	Yes	Yes	Yes	Yes
ChengxinI	None	www.ccxr.com.cn	Shanghai	Yes	Yes	No	No	No
Jincheng	None	www.dfratings.com	Beijing	Yes	Yes	Yes	Yes	Yes
Lianhe	None	www.lianhecreditrating.com.cn	Beijing	Yes	Yes	Yes	No	No
Pengyuan	None	www.pyrating.cn	Shenzhen	No	Yes	Yes	No	No
FarEast	None	www.sfecr.com	Shanghai	Yes	Yes	Yes	Yes	Yes

Notes: This Table describes the nine major Chinese CRAs in this study. The last five column reports the regulatory recognition of each CRA by various regulatory bodies in China.
NAFMII: National Association of Financial Markets Institutional Investors (regulator of the interbank market).
CIRC: China Insurance Regulatory Commission (Chinese insurance industry regulator).

[12] We obtain the regulatory approval information from the websites of the CRAs and cross-check with the approved CRAs lists from the regulatory agencies whenever available.

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